EE3600 – Practical 2: DC Motor Speed Control

Required speed task: pages 1–3 | Optional position: page 4

0. Purpose

Use your validated P1 values $\overline{K}$, $\overline{\tau}$ and gearbox ratio N for feedback speed control. Choose **one applicable method**: analytical, Ziegler–Nichols or computational. Simulate, implement on the Pico and assess the measured response. PI is sufficient if it meets the targets. Position control is optional (page 4).

1. Handover from Practical 1

Carry forward $G_{P1}(s) = \overline{K}/(\overline{\tau}s + 1)$, from terminal volts to gearbox-output rad/s. With signed fractional duty u ($0.40 = 40\%$) and motor-board supply V_s, use nominal $V_a \approx V_s u$:

$$G_n(s) = \frac{n_{\text{motor}}(s)}{u(s)} = \frac{K_{\text{PWM}}}{\tau s + 1}, \qquad \boxed{K_{\text{PWM}} = \overline{K}\, N\, \frac{60}{2\pi} V_s, \quad \tau = \overline{\tau}.}$$

K_{PWM} is motor RPM per fractional duty. The webpage's model helper labels this plant gain K_u; this sheet uses K_{PWM} to distinguish it from the Ziegler–Nichols ultimate gain.

- **Motor-drive change:** with power off, move the motor leads to the driver's **Motor 1** and connect the supply to the board's power input. Retain the P1 encoder wiring.
- **P2 settings: Counts / motor revolution = 44** (x4-decoded counts); **Gear ratio (motor / output) = 21.3** for the demonstrated platform; use the confirmed N for other kits. Do not enter P1's effective **937 output-shaft counts** as motor-shaft counts.

Keep the P1 motor and load; assess driver/supply effects in the new closed-loop results.

2. Speed Targets and Proportional Baseline

A) Required performance

Use an achievable step $r_0 \to r_f$ in motor RPM; compare designs under the same conditions.

Quantity	Target and definition
Overshoot	$\leq 5\%$ of the commanded step magnitude, beyond the final target.
Rise time t_r	≤ 0.50 s; 10–90% of the commanded change.
Settling time t_s	≤ 1.00 s; remain inside the target band for all later assessment samples, including at least 2 s final dwell.
Final tracking error	Last-2-s mean absolute error $\leq \max(0.02\lvert r_f\rvert, 2q_n)$ motor RPM. Also report signed bias.

For C_m counts per motor revolution and the 50 ms speed window, $q_n = 60/(0.050 C_m)$ motor RPM. The settling band is $b_n = \max(0.02\lvert r_f - r_0\rvert, 2q_n)$ about r_f. With $C_m = 44$, the floor is 54.5 motor RPM. Report the displayed band and whether the floor governs it. Insufficient data cannot establish a pass.

B) Simulate a P-control baseline

Use unity negative feedback, $e = r - y$, and $u = C(s)e$. For $C(s) = K_p$,

$$T(s) = \frac{C(s)G_n(s)}{1 + C(s)G_n(s)}, \qquad T_P(s) = \frac{K_{\text{PWM}} K_p}{\tau s + 1 + K_{\text{PWM}} K_p}.$$

The ideal $\tau_{\text{cl}} = \tau/(1 + K_{\text{PWM}} K_p)$ and step-error ratio $e_{ss}/r_f = 1/(1 + K_{\text{PWM}} K_p)$ show that finite P speeds the response and reduces, but does not remove, offset. Simulate and record a modest gain within the duty limit.

3. Analytical Method: Initial Speed PI Design

Choose a desired closed-loop time constant T_c. To obtain $T_n(s) = 1/(T_c s + 1)$ in the ideal model, use

$$C(s) = K_p + \frac{K_i}{s} = \frac{\tau s + 1}{K_{\mathrm{PWM}} T_c s}, \qquad \boxed{K_p = \frac{\tau}{K_{\mathrm{PWM}} T_c}, \quad K_i = \frac{1}{K_{\mathrm{PWM}} T_c}, \quad K_d = 0.}$$

The controller zero cancels the model's stable pole. This gives $t_r = \ln(9) T_c$, a 2% settling time $t_s = \ln(50) T_c$, zero nominal overshoot and zero nominal step error. Hence the required times give

$$T_c \le \min\left(\frac{0.50}{\ln 9}, \frac{1.00}{\ln 50}\right) = 0.228 \text{ s}.$$

For example, $T_c = 0.20$ s predicts $t_r = 0.439$ s and $t_s = 0.782$ s. Insert *your* K_{PWM} and τ to obtain initial gains. These are model predictions: finite duty, measurement delay and model mismatch may require a slower design or smaller reference step. Validate the complete response before implementation.

4. Empirical Method: Ziegler–Nichols

The ultimate-gain method needs a finite positive proportional gain K_u that produces sustained oscillation with period T_u. **The ideal first-order speed model has no such gain:** its characteristic polynomial $\tau s + 1 + K_{\mathrm{PWM}} K_p$ remains stable for every $K_p > 0$.

If the instructor supplies a suitable higher-order/delayed example or recorded experiment, identify K_u, T_u from it and use the table. Otherwise, explain the limitation and select the analytical or computational route. Do not keep increasing gain to force sustained oscillation on this model or the physical motor.

Controller	K_p	K_i	K_d
P	$0.5K_u$	0	0
PI	$0.45K_u$	$1.2K_p/T_u$	0
PID	$0.6K_u$	$2K_p/T_u$	$K_p T_u/8$

These parallel gains are initial values; Ziegler–Nichols may be too aggressive for a 5% overshoot target. The reaction-curve alternative also needs an applicable response with a resolved positive delay L; the delay-free first-order model has $L = 0$. For $Ke^{-Ls}/(Ts + 1)$, its PID rule is $K_p = 1.2T/(KL)$, including the plant gain.

5. Computational Method and Simulation Checks

Use MATLAB/Simulink or a suitable alternative with the converted speed model and unity negative feedback. Start with PI. Use a tuning tool or the lecture's parameter-search approach, state the search range/settings, and assess every candidate against the Section 2 targets.

- **Controller convention:** use parallel gains. If software reports $K_p(1 + 1/(T_i s) + T_d s)$, convert $K_i = K_p/T_i$ and $K_d = K_p T_d$. PI uses $K_d = 0$.
- **If $K_d > 0$:** match the Pico with PI on error minus filtered D on measured output, $T_f = 0.020$ s (derivative setpoint weight zero).
- Check reference tracking and applied duty, including output saturation. The control interval is 10 ms and the speed window is 50 ms. Do not add the same measurement lag twice if it was already included in the fitted P1 model.
- When using `stepinfo`, assess against the commanded step/final target, including any tracking offset; do not accept a response that merely settles about the wrong speed.

Whichever route you choose, save the initial gains and simulation, then use page 3 to test and refine the controller. You do not need to complete all three tuning methods on hardware.

6. Implement and Test Speed Control

Use the provided `EE3600_Practical_2_Pico.uf2` and the website's `practical2-control.html` page. Firmware installation follows the procedure already used in P1. The Pico runs the controller; the browser sets gains, records measurements and analyses the response.

1. Open the P2 page in Chrome/Edge and choose **Connect Pico**. In **Hardware setup**, enter the actual supply, page 1's encoder/gear settings and agreed duty, speed and trial limits. Choose **Apply configuration** while disarmed.
2. Enter $\overline{K}, \overline{\tau}$ with the output-rad/s-per-volt option; choose **Convert P1 model**, then **Use converted model**. If using the **Optional model-based gain helper**, enter T_c and choose **Calculate suggestions**. Copy reviewed gains into **Speed control**; suggestions never apply them automatically.
3. Enter **Target motor speed (RPM)** and K_p, K_i, K_d, leaving duty headroom. Keep $T_f = 0.020$ s initially. A 10 s trial accommodates the targets and final dwell. After changing to Motor 1, briefly check at low duty that positive duty gives positive speed.
4. Choose **Arm & run speed trial**. **STOP** disarms; allow for coasting. Afterwards choose **Analyse response** and **Save CSV**. Save before the next trial, which starts a new recording.
5. Stop before editing gains. Change one setting at a time and repeat under the same conditions. If the initial design passes, repeat it unchanged. If saturation, overshoot or noise grows, slow the tuning or reduce the demand.

K_p	K_i	K_d
Duty / motor RPM	Duty / (motor RPM · s)	Duty · s / motor RPM

Gains use **fractional duty**; the traces and CSV duty columns use percent. Do not copy gains from a percent-duty/output-RPM simulator without conversion. The firmware includes output limiting, anti-windup and filtered derivative on measurement. Reapply settings after a reboot.

7. Results, Discussion and Deliverables

Separate model predictions from measurements. Report initial and final gains with units and use this comparison table; add hardware repetitions if useful.

Case	Overshoot	t_r (s)	t_s (s)	Final error / bias
P baseline: simulation				
Initial design: simulation				
Initial hardware trial				
Final hardware trial / repeat				

Plot the reference, measured motor speed and applied duty for the initial and final trials. Use the armed trial data, excluding coasting after disarm. State the displayed settling/error bands, any saturation, and which targets were met. Discuss how the gain changes affected the response and why the motor differed from the prediction.

Submit the new P2 work:

- The carried-forward $\overline{K}, \overline{\tau}, N$, actual V_s, converted K_{PWM}, τ and P2 settings.
- Your chosen method, applicability, calculations or search settings, and P-baseline/controller simulations.
- Initial/final gains, measured P2 CSVs and plots, the comparison table and a short discussion. No P1 plots, CSVs or identification calculations need to be resubmitted.

Lecture reference: `EE3600-PID Controller.pdf`. Optional position work is entirely on the following page.

8. Optional Extension: Position Control

Complete the required speed task first. This extension controls the gearbox-output angle, in degrees relative to a stopped zero. Use the same tuning, simulation and measured-validation approach, with the position model and gains below.

A) Obtain the position model from the speed model

Because position integrates speed,

$$G_\theta(s) = \frac{\theta_{\text{out}}(s)}{u(s)} = \frac{A}{s(\tau s + 1)}, \qquad A = \frac{6K_{\text{PWM}}}{N} = \frac{180}{\pi} V_s \overline{K}.$$

This is output degrees per fractional duty; do not multiply by N again. With P control,

$$T_\theta(s) = \frac{AK_p}{\tau s^2 + s + AK_p}, \qquad \omega_n^2 = \frac{AK_p}{\tau}, \qquad 2\zeta\omega_n = \frac{1}{\tau}.$$

Thus **P cannot set damping and natural frequency independently**. The plant integrator gives zero nominal step-position error in a stable ideal loop; integral control is not automatically required. Test P first or proceed to PD/computational design.

B) Design PD from the desired response

Keep the lecture's optional **simulation** goals: overshoot $\leq 5\%$, $0.25 \leq t_s \leq 0.80$ s and zero nominal steady-state error. Choose damping and a settling-time aim; for the standard second-order approximation,

$$\zeta \geq 0.70 \quad \text{for a 5\% overshoot limit}, \qquad \omega_n \approx \frac{4}{\zeta t_s}.$$

For derivative on measurement, the ideal unfiltered controller is $u = K_p(r - \theta) - K_d\dot{\theta}$, giving

$$T_\theta(s) = \frac{AK_p}{\tau s^2 + (1 + AK_d)s + AK_p}, \qquad \boxed{K_p = \frac{\tau\omega_n^2}{A}, \quad K_d = \frac{2\zeta\omega_n\tau - 1}{A}, \quad K_i = 0.}$$

If $K_d < 0$, revisit the chosen response or test P; do not enter negative derivative gain. Validate the complete response with the 10 ms control interval, duty limits and filtered D on measurement ($T_f = 0.020$ s). PD on error adds a reference-path zero, so its step response is different.

The ideal position characteristic $\tau s^2 + s + AK_p$ is stable for every $K_p > 0$ and has no finite positive Ziegler–Nichols ultimate gain. Its integrator also rules out the usual open-loop reaction-curve rule. Use analytical/computational design unless a suitable different example is supplied.

C) Implement, measure and report

Open **Optional extension · Position control**. With the motor stopped, choose **Zero position**. Enter a **Target output angle (degrees)** within the configured travel limit and your position gains:

K_p	K_i	K_d
Duty / output degree	Duty / (output degree · s)	Duty · s / output degree

Choose **Arm & run position trial**, then **Analyse response** and **Save CSV**. The angle is relative to the chosen zero, without shortest-path wrapping. Encoder-derived position does not independently measure backlash or slip.

Report simulation goals separately from the browser's optional **hardware guide**: overshoot $\leq 5\%$, rise time ≤ 0.75 s, settling time ≤ 1.50 s and final mean absolute error $\leq b_\theta$, where $b_\theta = \max(2^\circ, 720/(C_m N)^\circ)$ is also the settling band. Require at least 2 s final dwell and use the last 2 s for error; $C_m = 44, N = 21.3$ gives $b_\theta = 2^\circ$. Add your model, gains, simulation, measured reference/angle/duty plots and numerical comparison to the report only if completing this extension.